

Data Structures Lab – Assignment 8

April 2, 2026

Assignment: Queue Operations, Deque, Circular Queues

1. Queue Implementation Using Array

Write a program to implement a **Queue** using an array. The program should support the following operations:

- Enqueue (insert an element)
- Dequeue (remove an element)
- Display the elements of the queue

The program must also handle **overflow and underflow conditions** appropriately.

Sample Input:

Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(), Display()

Sample Output:

Queue: 20 30

2. Palindrome Check Using Deque

Write a program using a Deque to determine whether a given string is a palindrome or not.

Sample Input:

String = madam

Sample Output:

The given string is a Palindrome

3. Application of Queue: Customer Service Simulation

In a bank, customers arrive and wait in a queue to be served. Each customer requires a certain amount of service time.

Given an array representing the **service time of each customer**, write a program to:

- Compute the **total waiting time** for all customers
- Compute the **average waiting time**

Sample Input:

$N = 4$

Service Time = [5, 3, 8, 6]

Sample Output:

Total Waiting Time = 31

Average Waiting Time = 7.75

4. Circular Queue Implementation

Write a program to implement a **Circular Queue** using an array of size N . The program should support the following operations:

- Enqueue
- Dequeue
- Display

Ensure that the queue efficiently utilizes space using **circular indexing** and correctly handles **overflow and underflow** conditions.

Sample Input:

$N = 5$

Operations:

Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(),
Enqueue(40), Enqueue(50), Enqueue(60), Display()

Sample Output:

Queue: 20 30 40 50 60